

Appendix A -- Chapter Mapping with Exercises

Volume #	Chapter	Exercise	Tutorial Section	Video	Notes
Volume-1	Chapter 00~03	N/A	N/A	N/A	Foundations & EKA Introduction
Volume-1	Chapter 04	Exercise 1, 2, 3	Section 4.0	04	Creating Named Classes
Volume-1	Chapter 05	Exercise 4	Section 4.1	05	Using a Reasoner
Volume-1	Chapter 06	Exercise 5	Section 4.2	06	Disjoint Classes
Volume-1	Chapter 07	Exercise 6	Section 4.3	07	Create Class Hierarchy
Volume-1	Chapter 08	N/A	N/A	N/A	Conceptual Modeling Foundations
Volume-2	Chapter 09	Exercise 7	Section 4.4	09	Object Properties
Volume-2	Chapter 09	Exercise 8	Section 4.5	09	Creating Property Hierarchies
Volume-2	Chapter 10	Exercise 9	Section 4.6	10	Inverse Properties
Volume-2	Chapter 11	Exercise 10	Section 4.7	11	Property Characteristics
Volume-2	Chapter 12	(Added by Author)	N/A	N/A	Property Characteristics Deep Dive

Volume #	Chapter	Exercise	Tutorial Section	Video	Notes
Volume-2	Chapter 13	Exercise 11, 12	Section 4.9	13	Domain & Range
Volume-3	Chapter 14	Exercise 13	Section 4.10.2	14	Existential Restrictions
Volume-3	Chapter 15	N/A	Section 4.10.2	14	Semantic Description (Theory)
Volume-3	Chapter 16	N/A	Section 4.10.2	14	Description Logic Foundations
Volume-4	Chapter 17	N/A	N/A	N/A	SKDL Introduction
Volume-4	Chapter 18	Exercise 14	Section 4.10.3	15	Stage 1 — Conceptual Modeling
Volume-4	Chapter 19	Exercise 15	Section 4.10.3	16	Stage 2 — Semantic Description
Volume-4	Chapter 20	Exercise 16	Section 4.10.3	17	Stage 3 — Knowledge Reuse (Cloning Pattern)
Volume-4	Chapter 20	Exercise 17	Section 4.10.3	18	Stage 3 — Knowledge Reuse (Practicing Specialization)
Volume-4	Chapter 21	Exercise 18	Section 4.10.3	19	Stage 4 — Semantic Governance (Disjointness)
Volume-4	Chapter 22	Exercise 19	Section 4.10.4	20	Stage 5 — Semantic Validation (Probe Class)
Volume-4	Chapter 23	Exercise 20	Section 4.11	21	Stage 6 — Semantic Reasoning (Primitive Classes)
Volume-4	Chapter 23	Exercise 21	Section 4.11		Stage 6 — Semantic Reasoning (Defined Classes)

Volume #	Chapter	Exercise	Tutorial Section	Video	Notes
Volume-4	Chapter 24	N/A	N/A		Stage 7 — Executable Knowledge (Introduction)
Volume-5	Chapter 25	Exercise 22	Section 4.12		Universal Restrictions
Volume-5		Exercise 23	Section 4.13		Open World Assumption & Closure Axioms
Volume-5		Exercise 24	Section 4.14		Enumerated Classes
Volume-5		Exercise 25	Section 4.15		hasValue Restrictions
Volume-5		Exercise 26	Section 4.15		Cardinality Restrictions
Volume-5		Exercise 27	Section 5.1		Datatype Properties
Volume-5		Exercise 28, 29, 30, 31, 32	Section 5.2		Data Property Values & Classification
Volume-5		Exercise 33	Section 9.1		DL Queries
Volume-5		Exercise 34, 35, 36	Section 10.0		SWRL & SQwRL Rules
Volume-6					Advanced Topics
Volume-7					EKA & AI Integration